UNIT 2 · FORCES & NEWTON'S LAWS · WEEK 4 · IN-CLASS REVIEW

Unit 2 Review — Systems & Inclined Planes

Physics 2026–27 · worked in class Tuesday, December 1 · Unit 2 Test Thu, Dec 3

NAME _____ DATE _____

How to work this review. This is **in-class review (CW), not homework** — work it with your table during Tuesday's block; it's your last practice before the test. Systems: use the *system view* (total mass, external forces only) to find a, then one object's *individual FBD* to find T. Inclines: tilt your axes with the ramp — parallel component $mg\ \sin\theta$, perpendicular $mg\ \cos\theta$, and $N = mg\ \cos\theta$, not mg. When in doubt, sanity-check your trig at $\theta = 0°$ and $90°$.

1 Two blocks, one string

SYSTEMS

A $6.0\ kg$ block and a $4.0\ kg$ block sit on a frictionless table, connected by a string; a $30\ N$ force pulls the 6.0 kg block (and the system) forward.

a. System view: find the acceleration.

b. Individual view: isolate the trailing 4.0 kg block and find the tension.

c. Check yourself on the 6.0 kg block: does $F - T = m_1 a$ hold with your numbers?

K	EVERY GIVEN, WITH UNITS
D	A FIRST, THEN T
E	SYSTEM: $F = (M_1+M_2)A$ · INDIVIDUAL: $T = M_2A$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

2 The Atwood machine

SYSTEMS

A 7.0 kg mass and a 5.0 kg mass hang from opposite ends of a string over a frictionless, massless pulley.

a. Which way does the system move? Say so before any algebra.

b. Find the acceleration. (The driving force is the *difference* in the weights.)

c. Find the tension. Then check: is T between the two weights? It has to be — say why.

K	EVERY GIVEN, WITH UNITS
D	A, THEN T
E	$(M_1-M_2)G = (M_1+M_2)A$ · THEN ONE MASS ALONE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

3 Frictionless ramp

INCLINES

A 3.0 kg block is released on a frictionless 25° incline.

a. Find its acceleration down the slope. (Notice what cancels.)

b. Find the normal force on the block.

c. A 30 kg block is released beside it. Which reaches the bottom first? Use your answer to (a).

K	EVERY GIVEN, WITH UNITS
D	THE TARGETS & THEIR UNITS
E	A = G SINθ · N = MG COSθ
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

4 Ramp with friction

INCLINES

A block slides down a 35° incline with $\mu_k = 0.30$. Find its acceleration. Mass isn't given — show why you don't need it.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	A = G(SINθ − M_k COSθ)
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Does it stay put?

STATIC CASE

A crate sits on a 15° loading ramp with $\mu_s = 0.30$. Does it slide on its own, or stay put? Use the tanθ vs. μ_s comparison from class — and show in one line of algebra where that comparison comes from.

WORK

6 The capstone combo

CHALLENGE

A `6.0 kg` block on a frictionless `30°` incline is connected by a string over a pulley to a `4.0 kg` mass hanging off the top edge.

a. Tug-of-war first: compare m_2g to $m_1g \sin\theta$. Which side wins, and which way does the system move?

b. Find the acceleration (system view — T is internal).

c. Find the tension (isolate the hanging mass).

K	EVERY GIVEN, WITH UNITS
D	A, THEN T
E	$M_2G - M_1G \sin\theta = (M_1+M_2)A$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

7 Sine or cosine?

SANITY CHECK

A classmate writes `N = mg sinθ` for a block on a ramp. Without touching a calculator, use the θ = 0° check (flat ground) to prove them wrong in two sentences.

ANSWER

*In-class review (CW) — not graded as homework. This is the last practice before the **Unit 2 Test (Thu, Dec 3)**; the combo in problem 6 is exactly the shape of the test's hardest item, so make sure your table can defend it at the board. Keep this sheet: it comes back in midterm review week (Dec 7–10), and the midterm covers Units 0–2.*